Science Olympiad — AddRidge - SSSS

Names of participants: (Please print neatly)

School Name:

Check your team: ☐ Varsity ☐ JV1 ☐ JV2 ☐ JV3

Warning: Do not open this packet until given permission to do so.

Note: There are useful notes after this page.

Scoring:

Time to solve first problem: _____ (use to calculate Bonus below)

Question	Value	Incorrect letters	Deduction	Score
Timed	200			
1	250			
2	200			
3	250			
4	350			
5	400			
6	450			
7	650			
8	550			
9	550			
10	500			
11	100			
12	100			
13	150			
14	250			
15	300			
16	300			
17	250			
18	250			
19	250			
20	300			
21	350			
22	250			
23	600			
Bonus				
Final Score				

The following tables might be useful during the event.

	A	**B**	**C**	**D**	**E**	**F**	**G**	**H**	**I**	**J**	**K**	**L**	**M**	**N**	**O**	**P**	**Q**	**R**	**S**	**T**	**U**	**V**	**W**	**X**	**Y**	**Z**
A	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
B	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A
C	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B
D	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C
E	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D
F	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E
G	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F
H	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G
I	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H
J	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I
K	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J
L	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K
M	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L
N	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M
O	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N
P	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
Q	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
R	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
S	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
T	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
U	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
V	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
W	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
X	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
Y	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
Z	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y

A	**B**	**C**	**D**	**E**	**F**	**G**	**H**	**I**	**J**	**K**	**L**	**M**	**N**	**O**	**P**	**Q**	**R**	**S**	**T**	**U**	**V**	**W**	**X**	**Y**	**Z**
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25

A	**B**	**C**	**D**	**E**	**F**	**G**	**H**	**I**	**J**	**K**	**L**	**M**	**N**	**O**	**P**	**Q**	**R**	**S**	**T**	**U**	**V**	**W**	**X**	**Y**	**Z**
Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A

1	3	5	7	9	11	15	17	19	21	23	25
1	9	21	15	3	19	7	23	11	5	17	25

AAAAA	A	**AABBA**	G	**ABBAA**	N	**BAABA**	T
AAAAB	B	**AABBB**	H	**ABBAB**	O	**BAABB**	U/V
AAABA	C	**ABAAA**	I/J	**ABBBA**	P	**BABAA**	W
AAABB	D	**ABAAB**	K	**ABBBB**	Q	**BABAB**	X
AABAA	E	**ABABA**	L	**BAAAA**	R	**BABBA**	Y
AABAB	F	**ABABB**	M	**BAAAB**	S	**BABBB**	Z

Frequency Table of English letters:

E -	12.51%	S -	6.54%	C -	3.06%	G -	1.96%	K -	0.67%
T -	9.25%	R -	6.12%	U -	2.71%	W -	1.92%	X -	0.19%
A -	8.04%	H -	5.49%	M -	2.53%	Y -	1.73%	J -	0.16%
O -	7.60%	L -	4.14%	F -	2.30%	B -	1.54%	Q -	0.11%
I -	7.26%	D -	3.99%	P -	2.00%	V -	0.99%	Z -	0.09%
N -	7.09%								

Frequency Table of Spanish letters:

E -	14.08%	I -	5.98%	M -	3.08%	Y -	1.09%	Z -	0.47%
A -	12.16%	L -	5.24%	P -	2.89%	V -	1.05%	Ñ -	0.17%
O -	9.20%	D -	4.67%	B -	1.49%	G -	1.00%	X -	0.14%
S -	7.20%	T -	4.60%	H -	1.18%	F -	0.69%	K -	0.11%
N -	6.83%	U -	4.69%	Q -	1.11%	J -	0.52%	W -	0.04%
R -	6.41%	C -	3.87%						

For the purposes of cryptograms it is customary to treat n and ñ as distinct letters, but a and á are the same letter. Likewise for e and é, and i and í. In other words, all the accent marks get amputated when working with cryptograms. Also, while some older Spanish dictionaries consider ch, ll, and rr, to be their own letters—this has fallen out of modern usage. Accordingly, “burro” is considered as five letters: “b-u-r-r-o” and not as four letters “b-u-rr-o.”

Morse Code:

A	●−
B	−●●●
C	−●−●
D	−●●
E	●

F	●●−●
G	−−●
H	●●●●
I	●●
J	●−−−

K	−●−
L	●−●●
M	−−
N	−●
O	−−−

P	●−−●
Q	−−●−
R	●−●
S	●●●
T	−

U	●●−
V	●●●−
W	●−−
X	−●●−
Y	−●−−
Z	−−●●

0	−−−−−
1	●−−−−

2	●●−−−
3	●●●−−

4	●●●●−
5	●●●●●

6	−●●●●
7	−−●●●

8	−−−●●
9	−−−−●

●	E
●●	I
●●●	S
●●●●	H
−●●●	B
−−−−−	0
●●●●●	5

−	T
●−	A
●●−	U
●●●−	V
−●●−	X
●−−−−	1
−●●●●	6

−●	N
●−●	R
●●−●	F
−●−●	C
●●−−−	2
−−●●●	7

−−	M
●−−	W
●−●●	L
−●−−	Y
●●●−−	3
−−−●●	8

−●●	D
●−−●	P
−−●●	Z
●●●●−	4
−−−−●	9

−●−	K
●−−−	J
−−●−	Q

−−●	G

−−−	O

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Timed Question **[200 points]** Solve this timed aristocrat. When you have solved it, raise your hand so that the time can be recorded and the solution checked.

LJ KJZZRHRC KOWR DJWR OIJVY NJU OKKVWOYRZT TJV KOS

OSCURW XVRCYPJSC OIJVY WOSLJD BOCCOHRC, JW UNOY TJV

KOS YWVZT LJ?

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency		1	7	2				2	2	12	6	3		2	12	1		8	5	4	3	7	7	1	6	4
Replacement																										

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1) **[250 points]** Solve this aristocrat where the ciphertext Z maps to plaintext D.

DXDKOTIZO UEGPDZ PI WGIU UVEP C UINJZ ZI CL C ZCZG'P

UCG. C ANDQQ UD'JJ GDXDK WGIU.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	1		6	7	2		6		6	3	2	1		2	2	4	2			1	7	1	2	2		6
Replacement																										

2) **[200 points]** Decode this quote about a cereal.

YL ADO CWKJQ CIB OPQYPE, Y ADYPT Y CWFJQ EKIX BWNO

RYPPINWP AWIBA RKFPRD, BWNO CIAOK, IPQ Y'Q SKWXIXJG

EW RKIVG.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	5	4	4	3	3	2	2		8	3	6	1		3	5	8	5	4	1	1		1	8	3	7	
Replacement																										

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3) **[250 points]** Solve this K1 aristocrat which is a quote from Nikola Tesla.

XNL OVLMNTW ZYU XNL CZVHVYDW ZLT VYWTJZLZAHT, HVGT

CXLBT ZYU IZMMTL. PETY METR WTJZLZMT, IZY VW YX IXLT.

K1	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	1	1	2	1	2		1	3	3	2		9	5	3	1	1		1		12	2	6	5	5	7	10
Replacement																										

4) **[350 points]** Solve this K2 aristocrat about chaos theory.

VFBPTPLBVFSYI, K APFODYDZ NRFLVLBVFS RQ BGR TROL VL

NRFLVOPTPO K OIFKZVN LILBPZ GVBU NUKRBVN LRYDBVRFL.

Replacement																										
K2	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	1	8		3		8	2		3		4	10		5	5	6	1	8	2	3	2	10			3	3

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5) **[400 points]** Solve this aristocrat that is a quote from comedian Jack Handey.

GSDRAS ZRW IAQOQIQUS KRYSRLS, ZRW KJRWPB FHPN H YQPS

QL OJSQA KJRSK. OJHO FHZ FJSL ZRW IAQOQIQUS OJSY,

ZRW HAS H YQPS HFHZ DARY OJSY HLB ZRW JHXS OJSQA

KJRSK.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	7	2		2		4	1	10	4	10	6	4		1	8	4	11	12	17		2		6	1	6	7
Replacement																										

6) **[450 points]** Solve this aristocrat that has been misspelled quite badly.

OYLOO ET EMKOB FOOFIG ZKENZ KEN MKOR UBO IYKUFFOO.

MKOQB XEYMQYIUG UMMOYZKIY ME EMKIBZ DOKULQEB ZKEZ

KEN MKOR UBO DENBQYH.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency		7		2	11	4	2	1	5		12	2	9	4	15		4	2		1	6			1	7	6
Replacement																										

7) **[650 points]** Solve this patristocrat.

FDQFW PDQJF WADFD YDFCW ACNCY HUJOC VHBVZ BHJXC

WUJHN YRDQF WEADQ BDHFE QDWCX BHJKC KYRBD JUMZA

COJDJ HHFMR JKJCK RJ

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	4	5	9	11	2	8		8		11	4		2	2	2	1	5	4			3	2	6	2	4	2
Replacement																										

8) **[550 points]** Solve this K1 Patristocrat that is a quote from inventor Dean Kamen.

KGKDO PAROT NJEAP KNPXK EABND EUQUB NEKUZ NSFEF

DKETN EETKO MNZEE AXUGK SADET NEJMT NEEKQ TZAXA

LOQNZ RA

K1	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	8	2		4	14	2	2			2	9	1	2	10	4	3	3	2	2	6	4			3		4
Replacement																										

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9) **[550 points]** Solve this patristocrat. You are told it starts with "Some".

NZLTZ JLKLZ NHSLW ZUHFR HLENS IFBTO WTUST RITNY

TUTJU ZLFME URHIG YSHDN ZRCNL KJUST RGGZY RBZFG

TGJZU LTFHF HTUUS MBTGS CSHFB AEFBS HK

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	1	5	2	1	3	8	6	9	3	4	3	8	2	6	1			6	9	12	9		2		3	9
Replacement																										

10) **[500 points]** Solve this K1 xenocrypt that has been encoded with an English keyword.

VWUXJBEILY DVAX IÑADLY VKLY MÑX HLY UEALY WÑXBXJ

NXJXU BEJXUL, WXUL HLY WLZUXY NEXJXJ NEXIWL.

K1	A	B	C	D	E	F	G	H	I	J	K	L	M	N	Ñ	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	3	3		2	5			2	3	6	1	10	1	3	3							6	3	5	13	7	1
Replacement																											

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11) **[100 points]** Solve this caesar cipher.

Q	N		Q		K	W	C	T	L		Z	M	U	M	U	J	M	Z		B	P	M		V	I	U	M	A		W	N		I	T	T

B	P	M	A	M		X	I	Z	B	Q	K	T	M	A	,		Q	'	L		J	M		I		J	W	B	I	V	Q	A	B	.
															,			'																	.

12) **[100 points]** Solve this Caesar cipher that is a quote about salad.

S	L	G	T	Y	R		N	L	P	D	L	C		D	L	W	L	O		H	T	E	S	Z	F	E		N	C	Z	F	E	Z	Y	D		T	D

W	T	V	P		S	L	G	T	Y	R		X	L	N		L	Y	O		N	S	P	P	D	P		H	T	E	S	Z	F	E		E	S	P

N	S	P	P	D	P	.
						.

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13) **[150 points]** Encode this phrase given that a = 15 and b = 7.

P	R	O	G	R	A	M	M	I	N	G		I	S	N	'	T		A	B	O	U	T		W	H	A	T		Y	O	U		K	N	O	W	;
															'																							;

I	T	'	S		A	B	O	U	T		W	H	A	T		Y	O	U		C	A	N		F	I	G	U	R	E		O	U	T
		'																															

14) **[250 points]** Solve this affine cipher that ends with "LS", and is a quote from Kawhi Leonard.

V	'	I	J		X	E	L	X	R	Z		E	V	B	J	G		S	N	H	J	-	D	N	N	B	J	G		H	J	X	E	Z	.
	'																					-														.

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15) **[300 points]** Encode this 3 by 3 Hill with keyword 'Aardvarks'.

$$\begin{pmatrix} A & A & R \\ D & V & A \\ R & K & S \end{pmatrix} \equiv \begin{pmatrix} 0 & 0 & 17 \\ 3 & 21 & 0 \\ 17 & 10 & 18 \end{pmatrix}$$

A	R	T	H	U	R	T	I	M	O	T	H	Y	R	E	A	D	

16) **[300 points]** Decode this hill cipher.

$$\begin{pmatrix} H & O \\ W & D \end{pmatrix} \equiv \begin{pmatrix} 7 & 14 \\ 22 & 3 \end{pmatrix}$$

I	L	Y	H	A	N	I	N	U	M

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17) **[250 points]** Decode this Vigenere cipher, whose keyword is 'Creed'.

K	M	I	F	H

G	E	M	R	Y

Q	C	Z	I	G

K	E	E	R	X

O	S	I	V	R

H	T	Y	P	W

U	S	S	X	K

C	C	I	E	G

G	I	E	R	G

C	W	S	P	O

Q	N	I	V	B

Q	L	L	E	Y

G	D	S	V	H

H	L	R	E	V

C	W	S	P	O

Q	N	I	V	E

W	K	C	S	X

O	R	O	I	P

Q	I	I	Q	R

P	V	C	E	V

C	C	I	E	G

G	I

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18) **[250 points]** Decode this Vigenere that starts with "Learn From"

P	M	N	J	G

J	Z	B	Q	G

R	K	M	I	Z

Q	E	G	Y	A

O	I	N	B	V

B	B	V	T	C

P	B	T	M	S

G	K	X	W	Z

S	Z	E	G	P

X	P	R	M	U

C	G	K	X	I

A	X	B	U	A

G	K	Q	F	R

W	G	L	H	W

B	B	T	Y	H

W	L	X	Q	B

R	Q	A	Y

19) **[250 points]** Solve this Morbit cipher where 1 = '••', 2 = --, 9 = 'x•', 8 = 'x-', 4 = '-•', and 5 = '•-'.

4 9 5 8 6 3 5 3 2 6 1 6 1 3 9 6 5 1 8 4 8 2 9 4 9 3

6 1 1 8 6 1 3 8 5 3 5 8 3 8 1 3 3 9 5 9 1 9 8 1 7 6

2 6 9 1 8 2 9 4 3 1 5 9 7 6 1 1 9 7 6 5 3 5 9 1 6

3 5 1 9 3 4 8 4 7 1 3 5 9 4 3 3 1 3 2 9 6 4 7 5 4

9 4 8 2 8 1 3 5 1 9 8 6

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20) **[300 points]** Decipher this quote encoded using the Morbit cipher. You are told it ends with "CRAZY" and that 6 maps to '-x'.

7 9 5 6 2 6 7 6 5 7 8 7 9 7 8 6 5 8 4 1 3 4 1 9 6

7 7 1 6 6 1 3 7 4 9 4 2 8 2 1 5 9 4 4 8 2 1 5 1 4 3

4 3 1 5 8 3 4 3 3 5 3 7 3 6 5 8 3 4 3 8 6 2 6 5 3

7 3 3 8 5 3 2 6 4 3 3 1 6 8 1 6 5 4 9 4 3 5 8 3 4 4

1 1 4 9 5 6 2 6 7 6 5 7 8 7 9 4 3 5 3 7 1 7 6 3 8 5 6

2 6 7 6 8 5 3 5 3 5 8 4 3 4 2

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21) **[350 points]** Decode this Pollux cipher that is a quote from Tyler, The Creator given that 1 = x, 2 = •, 5 = -, 7 = -, 9 = x, 0 = x

22693280555932903668824685122251890870051428494947 4185

6833498802481309242973310289988304849260750461230422 6053

6347311333983303211598284180272085947341880844030 0873

9444898152614291379953625672598805417721651428412 1153

9487022418867838

22) **[250 points]** Solve this tongue twister encoded as a Baconian cipher.

^!!!^!^!!!^!^!^^!!!^!^!^!!!^!!!!^!!!^!!^^!!!^^!^!!!!!

!!!^^!!^!!!^^!!!^^!^!!!!!!!!!^!^^^!!!^^!^!!!^!!!!!^!^^

!!^!!^!^!^!^^!^!!!^!^^!^^!!^^^!!^!!!^^^^!^!!!!!!!^!!!

!!!!^^^!!!^

23) **[600 points]** Megan, has faithfully followed the steps of the RSA key-generation algorithm. Here are the results:

$p = 341$

$q = 623$

$n = 232481$

$\Phi = 341271$

$e = 235663$

Unfortunately, Megan doesn't know how to compute the value of *d* and needs you to do that final step for them.

Enter the computed value of *d*, NOT the formula.